CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

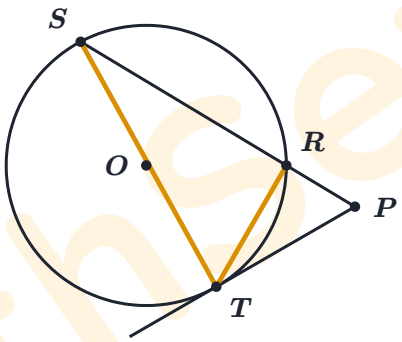
Chapter 18: Tangents and Intersecting Chords

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Tangent and secant from an external point

1. PT is a tangent to a circle at T, and a line through P cuts the circle at R and S, with R nearer to P. (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT. ICSE 2025, 3 marks



SOLUTION

(a) In $\triangle PTR$ and $\triangle PST$: $\angle P$ is common, and $\angle PTR = \angle PST$ (angle between tangent and chord equals the angle in the alternate segment). So $\triangle PTR \sim \triangle PST$ (AA).

(b) Corresponding sides are proportional: $\frac{PT}{PS} = \frac{PR}{PT}$, so $PT^2 = PR \times PS$.

(c) $PT^2 = 4 \times 16 = 64$

Answer: $PT = 8$ **cm**

2. A circle has centre O and radius 9 cm. The diameter AB is produced to a point P, and PQ is a tangent from P. If $PA = 24$ cm, find PQ. ICSE 2024, 3 marks

SOLUTION

$AB = 18$ cm, so $PB = PA - AB = 24 - 18 = 6$ cm.

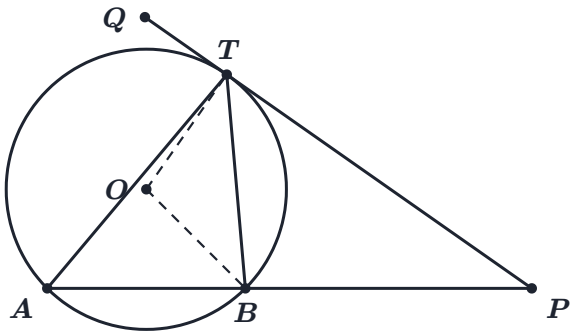
$PQ^2 = PA \times PB = 24 \times 6 = 144$

Answer: $PQ = 12$ **cm**

(Check: $OP = 15$ cm and $OQ = 9$ cm, so $PQ = \sqrt{15^2 - 9^2} = 12$ cm.)

3. PQ is a tangent to a circle with centre O at T. The chord AB produced meets the tangent at P. $AB = 9$ cm, $BP = 16$ cm, $\angle PTB = 50°$ and $\angle OBA = 45°$. Find (a) PT (b) $\angle BAT$ (c) $\angle BOT$ (d) $\angle ABT$.

ICSE 2023, 4 marks



SOLUTION

(a) $PA = 9 + 16 = 25$ cm. $PT^2 = PB \times PA = 16 \times 25 = 400$, so $PT = 20$ cm.

(b) By the alternate segment theorem, $\angle BAT = \angle PTB = 50°$.

(c) Angle at the centre is twice the angle at the circumference: $\angle BOT = 2 \times 50° = 100°$.

(d) $OB = OT$, so $\angle OBT = \dfrac{180° - 100°}{2} = 40°$. $\angle ABT = \angle OBA + \angle OBT = 45° + 40° = 85°$.

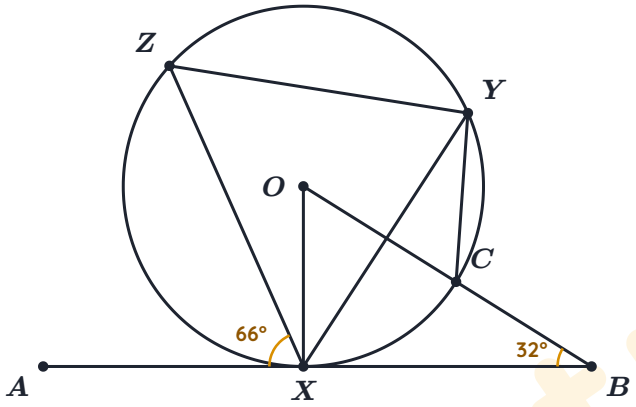
Answer: (a) 20 cm (b) $50°$ (c) $100°$ (d) $85°$

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2 The alternate segment theorem

4. X, Y, Z and C lie on a circle with centre O. AB is the tangent at X, OB cuts the circle at C, and $ZY = XY$. Given $\angle OBX = 32°$ and $\angle AXZ = 66°$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$.

ICSE 2025, 4 marks



SOLUTION

(a) $OX \perp AB$ (radius and tangent). In $\triangle OXB$: $\angle BOX = 90° - 32° = 58°$.

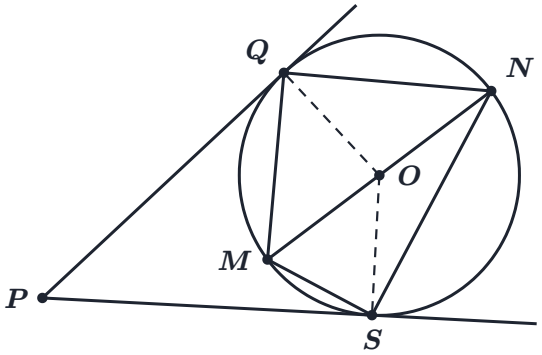
(b) The arc CX subtends $\angle COX = 58°$ at the centre and $\angle CYX$ on the circle: $\angle CYX = \frac{58°}{2} = 29°$.

(c) By the alternate segment theorem, $\angle ZYX = \angle AXZ = 66°$.

(d) $ZY = XY$, so $\angle YXZ = \frac{180° - 66°}{2} = 57°$. Also $\angle OXZ = 90° - 66° = 24°$. So $\angle OXY = 57° - 24° = 33°$.

Answer: (a) $58°$ **(b)** $29°$ **(c)** $66°$ **(d)** $33°$

5. PR and PT are tangents from P to a circle with centre O, touching it at Q and S. MN is a diameter, with M on the arc QS nearer to P. Given $\angle PQM = 42^\circ$ and $\angle PSM = 25^\circ$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. ICSE 2024, 4 marks



SOLUTION

(a) $OQ \perp PR$ (radius and tangent), so $\angle OQM = 90^\circ - 42^\circ = 48^\circ$.

(b) By the alternate segment theorem, $\angle QNM = \angle PQM = 42^\circ$ and $\angle SNM = \angle PSM = 25^\circ$. So $\angle QNS = 42^\circ + 25^\circ = 67^\circ$.

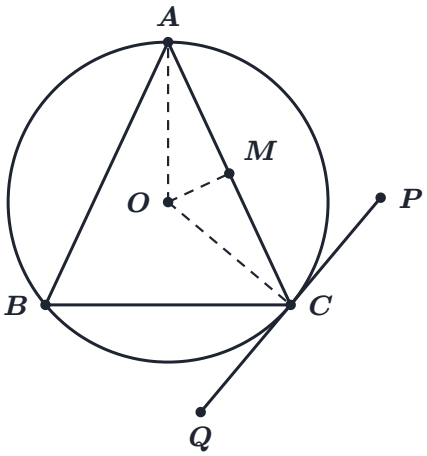
(c) Angle at the centre is twice the angle at the circumference: $\angle QOS = 2 \times 67^\circ = 134^\circ$.

(d) $QNSM$ is a cyclic quadrilateral: $\angle QMS = 180^\circ - 67^\circ = 113^\circ$.

Answer: (a) 48° **(b)** 67° **(c)** 134° **(d)** 113°

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6. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O. PQ is the tangent at C, with Q on the side of BC away from A. OM is perpendicular to the chord AC and $\angle COM = 65°$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. ICSE 2024, 3 marks



SOLUTION

(a) The perpendicular from the centre bisects the chord and the angle at the centre: $\angle AOC = 2 \times 65° = 130°$. So $\angle ABC = \frac{130°}{2} = 65°$.

(b) $AB = AC$, so $\angle ACB = \angle ABC = 65°$ and $\angle BAC = 180° - 130° = 50°$.

(c) By the alternate segment theorem, $\angle BCQ = \angle BAC = 50°$.

Answer: (a) $65°$ **(b)** $50°$ **(c)** $50°$

7. RT is a tangent to a circle at S. P and Q are points on the circle with $\angle PST = 30°$ and $\angle SPQ = 60°$. Find $\angle PSQ$. (a) $40°$ (b) $30°$ (c) $60°$ (d) $90°$ ICSE 2023, 1 mark

SOLUTION

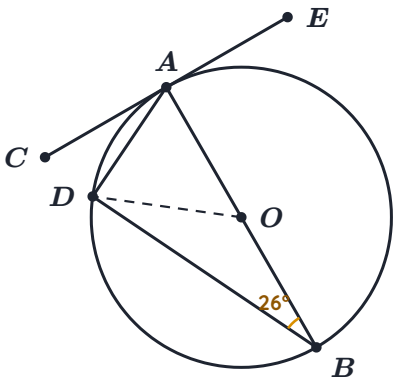
By the alternate segment theorem, $\angle PQS = \angle PST = 30°$.

In $\triangle PQS$: $\angle PSQ = 180° - 60° - 30° = 90°$

Answer: (d)

8. AB is a diameter of a circle with centre O, and CE is the tangent at A. D is a point on the circle on the same side of AB as C, with $\angle ABD = 26°$. Find (a) $\angle BDA$ (b) $\angle BAD$ (c) $\angle CAD$ (d) $\angle ODB$.

ICSE 2023, 4 marks



SOLUTION

(a) $\angle BDA = 90°$ (angle in a semicircle).

(b) $\angle BAD = 180° - 90° - 26° = 64°$

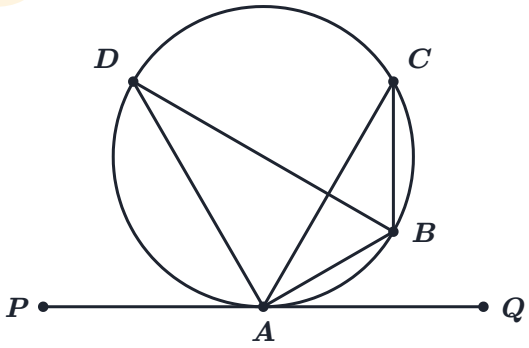
(c) By the alternate segment theorem, $\angle CAD = \angle ABD = 26°$.

(d) $OD = OB$ (radii), so $\angle ODB = \angle OBD = 26°$.

Answer: (a) $90°$ **(b)** $64°$ **(c)** $26°$ **(d)** $26°$

9. PQ is a tangent to a circle at A. B, C and D are points on the circle such that AB bisects $\angle CAQ$ and AD bisects $\angle PAC$. If $\angle BAQ = 30°$, prove that (a) BD is a diameter (b) $\triangle ABC$ is isosceles.

ICSE 2017, 3 marks



SOLUTION

(a) $\angle CAQ = 2 \times 30° = 60°$, so $\angle BAC = 30°$ and $\angle PAC = 180° - 60° = 120°$.

AD bisects $\angle PAC$, so $\angle CAD = 60°$. Then $\angle BAD = \angle BAC + \angle CAD = 30° + 60° = 90°$.

A chord that subtends a right angle at the circumference is a diameter, so BD is a diameter.

(b) By the alternate segment theorem, $\angle ACB = \angle BAQ = 30°$. Also $\angle BAC = 30°$.

Two angles of $\triangle ABC$ are equal, so $AB = BC$. **Both results are proved.**

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3 Tangent properties

10. PS and PT are tangents to a circle from P, with $\angle SPT = 80°$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) $140°$ (b) $90°$ (c) $80°$ (d) $50°$ ICSE 2024, 1 mark

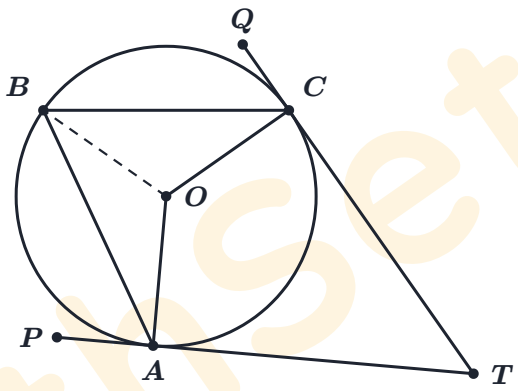
SOLUTION

$PS = PT$, so $\angle PTS = \angle PST = \dfrac{180° - 80°}{2} = 50°$.

$SQ \parallel PT$ and ST is a transversal, so $\angle QST = \angle PTS = 50°$ (alternate angles).

Answer: (d)

11. TP and TQ are tangents to a circle with centre O, touching it at A and C. B is a point on the circle with $\angle BCQ = 55°$ and $\angle BAP = 60°$. Find (a) $\angle OBA$ and $\angle OBC$ (b) $\angle AOC$ (c) $\angle ATC$. ICSE 2020, 4 marks



SOLUTION

(a) $OA \perp TP$, so $\angle OAB = 90° - 60° = 30°$. $OA = OB$, so $\angle OBA = 30°$.

$OC \perp TQ$, so $\angle OCB = 90° - 55° = 35°$. $OB = OC$, so $\angle OBC = 35°$.

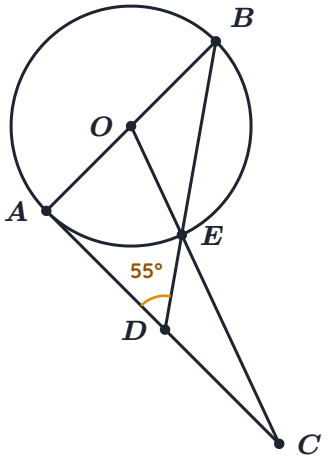
(b) $\angle ABC = 30° + 35° = 65°$, so $\angle AOC = 2 \times 65° = 130°$.

(c) In quadrilateral $OATC$ the angles at A and C are $90°$, so $\angle ATC = 180° - 130° = 50°$.

Answer: (a) $30°$ **and** $35°$ **(b)** $130°$ **(c)** $50°$

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12. AB is a diameter of a circle with centre O, and AC is the tangent at A. D is a point on AC, and BD cuts the circle at E. The line OE produced meets the tangent at C. If $\angle ADB = 55°$, find $x = \angle ACO$ and $y = \angle AOE$. ICSE 2019, 3 marks



SOLUTION

$\angle BAD = 90°$ (the tangent is perpendicular to the radius at the point of contact).

In $\triangle ABD$: $\angle ABD = 180° - 90° - 55° = 35°$.

$y = \angle AOE = 2 \times \angle ABE = 70°$ (the angle at the centre is twice the angle at the circumference).

In $\triangle OAC$: $x = 180° - 90° - 70° = 20°$.

Answer: $x = 20°$, $y = 70°$

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